

Temperature-sensitive poly-NIPAm modified cellulose nanofibril cryogel microspheres for controlled drug release

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Abstract In this article, we report a novel method for synthesizing temperature-sensitive polymer-modified cellulose nanofibril (CNF) cryogel microspheres. The pristine cryogel microspheres were first prepared using a spray-freeze dry method in the presence of a chemical crosslinker. Afterwards, NIPAm (N-isopropylacrylamide), a temperature-sensitive monomer, was polymerized and grafted to the cellulose cryogel microspheres through in situ free radical polymerization in the cryogel microreactor. The morphology, chemical structure, thermal sensitivity, bulk density and water uptake capacity of the hybrid microspheres were characterized. The CNF–PNIPAm hybrid microspheres exhibited a good temperature response at around 32 °C in water. The swelling behavior and drug release capability of CNF–PNIPAm hybrid microspheres were also investigated. The microspheres with PNIPAm exhibited a controllable drug release rate. The temperature effect on the drug release rate was also observed. These results indicated that porous CNF–PNIPAm hybrid microspheres could

serve as a new type of material for controlled drug release.

Keywords Cryogel microspheres · Nanofibrillar cellulose · Poly(N-isopropylacrylamide) · Drug delivery system

Introduction

Recently, the use of porous materials as a matrix for drug loading and release systems has raised research interest. Aerogel is a 3D network material providing an ultra-light weight with outstanding surface area and up to 99 % porosity, which is suitable for drug loading and drug release control. Many inorganic aerogels, such as silica aerogels, exhibit high drug loading in drug delivery (Guenther et al. 2008; Smirnova et al. 2003). However, the nonbiodegradable property of inorganic aerogels limits their use for drug delivery. The use of natural polysaccharide aerogels and their derivatives, such as alginate and starch aerogels, are especially attractive because of their biodegradability and biocompatibility (Ulker and Erkey 2014; Mehling et al. 2009; García-González et al. 2011; Mikkonen et al. 2013).

Cellulose is the most abundant biopolymer on earth. Moreover, it is safe, stable, nontoxic and biodegradable in nature. With abundant hydroxyls, cellulose can be easily converted into different functional materials, making it a promising candidate as a scaffold and

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carrier for tissue engineering and drug delivery (Novotna et al. 2013). As the new generation of materials, cellulose aerogel has received much attention in applications such as oil removal (Sai et al. 2015), waste water treatment (Zhang et al. 2012), supercapacitors (Zhang et al. 2014) and many others. The traditional method to prepare cellulose aerogel undergoes a sol-gel process (Gavillon and Budtova 2007; Innerlohinger et al. 2006). Toxic organic solvents of regenerated cellulose and the conventional cellulose crosslinkers such as N-methylmorpholine-N-oxide calcium and thiocyanate make them not suitable for use in medical applications. Cellulose nanofibrils with lengths of dozens of microns can be isolated from wood fibers by a simple mechanical shearing process, which exhibits a low density, high specific strength and modulus, and large surface area. Owing to these unique properties, CNF aerogels exhibit higher strength and are more robust with higher microstructure properties compared with regenerated cellulose aerogels. Nontoxic and robust CNF aerogel microspheres used as a cell culture scaffold prepared by spraying a CNF suspension into liquid nitrogen followed by freeze-drying were reported by Cai et al. (2014). The spherical structure of the aerogel has been widely used as a drug delivery matrix because it exhibits special properties that are substantially different from those of bulk material (Song et al. 2010).

Composite materials are always attractive because of the combination of diverse functionalities in one entity. The performance of aerogels as carriers can be improved by using the composite method. Many reports about hybrid aerogels show their outstanding physicochemical properties, these hybrid aerogels possess both the original properties of aerogels (porous structure and large surface area) and the modifier's functionalities (Ayers and Hunt 2001; Litschauer et al. 2011). The silica-chitosan hybrid aerogel system was investigated by Molvinger et al. (2004). The hybrid aerogels showed significantly different textural properties compared with chitosan aerogel, and a large amount of gentamicin (17 wt%) was loaded. Kadib et al. (2008) synthesized chitosan-titanium's hybrid aerogel for drug delivery and showed improved textural properties and stable mechanical properties. The environmentally sensitive thermoresponsive poly(N-isopropylacrylamide) (PNIPAAm) exhibits a lower critical solution temperature (LCST) in the range of 30–32 °C. The LCST temperature of

PNIPAM is close to the human body temperature, which makes it a very attractive candidate for making stimuli-responsive drug delivery materials (You et al. 2008). However, the fatal weakness of PNIPAM is evident: weak strength and rapid dissolution in water. Therefore, incorporation of PNIPAM into the cellulose matrix has been reported as an effective way to prepare smart materials. Hebeish et al. (2014) prepared thermoresponsive hydrogels through concurrent free radical polymerization of NIPAM and crosslinking BMA along with cellulose nanowhiskers.

In this study, we synthesized CNF–PNIPAM hybrid microspheres by using CNF cryogel microspheres as a microreactor. NIPAM monomer was polymerized and crosslinked in the interior of CNF cryogel microspheres. Owing to the highly porous structure and excellent water uptake capacity, the hybrid microsphere exhibited a high drug-loading capacity and unique release properties.

Experimental

Materials

KymeneTM (polyamide-epichlorohydrin resin) was received from Ashland Hercules, Inc., USA. Commercial bleached kraft southern pine pulp from a southeast US company was used as the raw material. N-isopropylacrylamide (NIPAm, 97 %) and hexane (95 %) were purchased from VWR. NIPAm was recrystallized from hexane and dried under vacuum prior to use. N,N'-methylenebisacrylamide (BMA) and potassium persulfate (KPS) were all laboratory-grade chemicals. 5-Fluorouracil (5-FU, ≥ 99 %) was purchased from Sigma.

Preparation of crosslinked cryogel microspheres

Commercial bleached softwood kraft pulp was used as the starting material for preparing CNF and original cellulose fibers. Dry cellulose pulp boards were soaked in deionized water for 24 h and then dispersed using a mechanical blender for 30 min to obtain a consistent fiber suspension. The native cellulose fiber was obtained. After that, the cellulose fiber suspension was subjected to the mechanical defibrillation process using a high-speed shear homogenizer (T18 basic, Ultra Turrax, IKA Works, Inc., USA) at 20,000 rpm

for 3 h to obtain the CNF. The preparation method for crosslinked cryogel microspheres was detailed in the study by Hongli Cai et al. (2014). First, a weighted amount of crosslinker, KymeneTM, was added to the cellulose suspension with mechanical stirring and finally treated in an ultrasonic bath for 10 min to obtain the native cellulose nanofibril aqueous suspension. Then the suspension was transferred to a steel feed tank, and the cellulose was sprayed and atomized at 40-MPa constant pressure through a 1.0-mm inner steel nozzle into liquid nitrogen. Finally, the frozen samples were freeze-dried in the freeze-drier (VirTis Freezemobile 25EL Sentry 2.0, USA) at a condenser temperature of $-84.5\text{ }^{\circ}\text{C}$ for 3 days. The crosslinking of the cryogel microspheres was achieved by curing the dried samples containing the crosslinker in a vacuum oven at $120\text{ }^{\circ}\text{C}$ for 3 h.

Preparation of CNF–PNIPAm hybrid microspheres

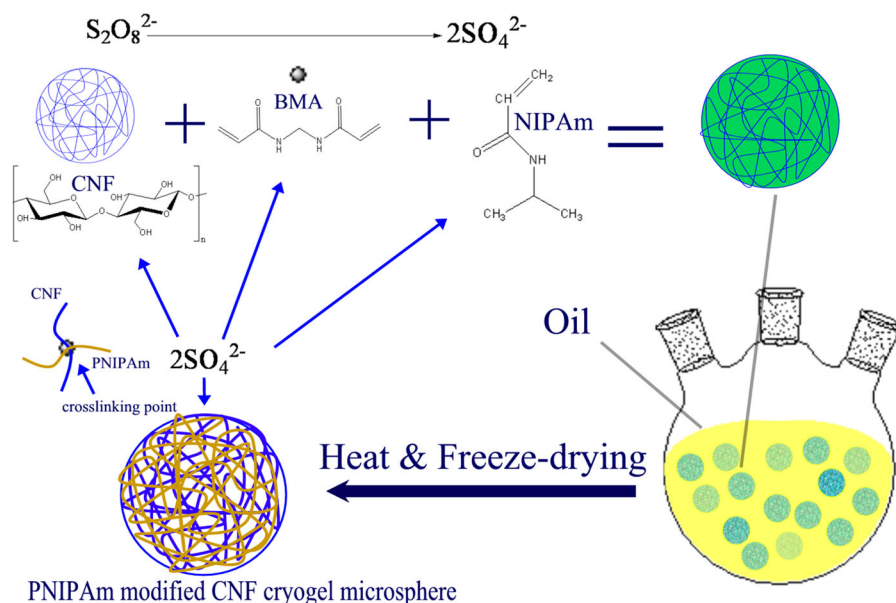
The CNF–PNIPAm hybrid composites were prepared by concurrent crosslinking and free radical polymerization of NIPAm inside the CNF cryogel microspheres as described in Scheme 1. First, the monomer N-isopropyl acrylamide, crosslinking agent N,N'-methylenebisacrylamide (BMA) and initiator potassium persulfate (KPS) were dissolved in distilled water to make a uniform solution, and the concentrations of

NIPAm monomer were controlled at 6, 9, 12 and 15 % (w/v), respectively. The BMA and KPS concentrations were 3 and 2 % based on the weight of NIPAm monomer. Afterwards, 1 g of dry CNF cryogel microspheres was soaked in the mixed solution. After gently shaking, the CNF cryogel microspheres were fully saturated by absorbing the solution containing the monomer (NIPAm), crosslinker (BMA) and initiator (KPS). The excess solution was wiped with blotting paper, then the sample was transferred into a three-necked bottle; kerosene and the emulsifier span-85 were added to the bottle at the same time. The microspheres saturated with monomer aqueous solution were dispersed in kerosene oil by vigorous shaking and heated to $70\text{ }^{\circ}\text{C}$ in a water bath under the protection of nitrogen. After 2 h polymerization, CNF–PNIPAm microspheres were washed by excessive acetone and distilled water several times. The purified sample was frozen for further drying by a freeze-drier. The corresponding samples were named AM-1, AM-2, AM-3 and AM-4 according to the concentrations of NIPAm monomer during the preparation process, and the CNF microsphere was named AM-0.

Preparation of drug (5-fluorouracil)-loaded microspheres

Model drug of 5-FU loading on the following samples: AM-0, AM-1, AM-2, AM-3 and AM-4 microspheres

Scheme. 1 In situ polymerization of NIPAm/crosslinker in the microreactor (monomer solution-saturated cryogel microspheres dispersed in kerosene)



were prepared in 8 mg/ml 5-FU. The mixture was stirred at 22 °C for 48 h. After that, the microspheres were collected and quickly washed (filtration method) by a certain amount of distilled water to remove the free drug on the surface of the microspheres. Each sample was washed by 500 ml distilled water, and water in the drug-loaded microspheres was then removed by the freeze-drying method.

To evaluate the 5-FU content in microspheres, dried drug-loaded microspheres were weighed. Drug contents were calculated by the following equation:

$$\text{Drug}_{\text{content}}(\text{wt}\%) = \frac{m_{\text{drug}}}{m_{\text{drug-loaded}}} \times 100$$

(m_{drug} : the weight of 5-FU; $m_{\text{drug-loaded}}$: the weight of 5-FU loaded microspheres)

In vitro drug release

The drug release behavior of 5-FU from CNF–PNIPAm hybrid microspheres was investigated in phosphate-buffered saline (PBS) solution at pH 7.2. The temperature-dependent drug release experiments were carried out with a horizontal laboratory shaker at 22 and 37 °C. Accurate quantities of the dry samples (0.1 g) were added to PBS. The concentration of 5-FU released was determined by the UV spectrophotometric method at 271 nm at the specific time. The fresh PBS was added back to keep the total volume constant. Each release experiment was repeated three times. The reference curve was fitted by $A = 0.05562C - 0.00171$, $R^2 = 0.99946$. (C: 5-FU concentration).

Characterization and analyses

The dried powdery samples were analyzed by Fourier transform infrared spectroscopy (FTIR). The spectra were recorded using Nicolet Magna-IR 550 equipment with a detector at 4 cm^{-1} resolution from 600 to 4000 cm^{-1} and 32 scans per sample.

Thermal stability of the CNF–PNIPAm was tested by thermogravimetric analysis (TGA, PerkinElmer STA 6000 thermal analyzer). Approximately 10 mg powder of the sample was used for each testing. The experiment was done at a heating rate of 10 °C min^{-1} under nitrogen flow of 20 ml min^{-1} .

Scanning electron microscopy (SEM, Zeiss LEO 1530 microscopy) was used for morphological

characterization. Samples were attached to the holders with conductive double-sided carbon tape and sputter coated with gold to avoid charging during the tests.

Bulk density was determined by transferring a known quantity of microspheres to a 100-ml measuring cylinder and tapping it three times. The bulk density of the cryogel was determined by dividing the weight of the sample by the volume of the plastic centrifuge tube.

Particle size of CNF and CNF–PNIPAm modified cryogel microspheres was determined by averaging more than 100 particles using an optical microscope (Olympus, India).

Angle of repose of different microspheres was determined by a fixed funnel method (Ritschel et al. 1980). The microspheres were poured into a funnel fixed at a position so that the outlet orifice of the funnel was 10 cm above a level black surface. The microspheres flowed down from the funnel to form a cone on the surface, and the angle of repose was then calculated by measuring the height of the cone and diameter of its base. Every sample was repeated three times.

Results

Preparation and morphology of CNF–PNIPAm cryogel microspheres

Softwood pulp fibers with diameters between 20 and 50 μm were used as the starting material to produce CNF. During the high-speed shear homogenization process, the CNFs were gradually isolated from the macroscopic pulp fibers because of the repeated shear forces. After 120 min, the CNFs with diameters mostly ranging from 10 to 40 nm were finally obtained; however, some large-sized CNFs were also found. The gel-like CNF suspension with a concentration of 1 % (w/w) was used to prepare CNF cryogel microspheres using the spray method. In order to improve the mechanical strength, particularly the wet strength of the microspheres, 5 wt% crosslinker KymeneTM (based on CNF) was added to the suspension with constant stirring. The pressure was set to 40 MPa during the spraying process. Once the atomized droplets had been frozen in liquid nitrogen, the frozen spherical sample was freeze-dried to obtain cryogel microspheres. The dried cryogel microspheres

were cured in a vacuum oven at 120 °C for 3 h to achieve sufficient covalent crosslinking.

CNF–PNIPAm hybrid microspheres were synthesized inside the CNF microspheres by free radical polymerization of NIPAm monomer in the presence of the BMA crosslinking agent using KPS as an initiator. The surfactant Span-85 was used to keep the polymerization entirely inside the microspheres. The NIPAm solution was locked inside the CNF microspheres during the polymerization process along with constant stirring. As a result, the microspheres are almost spherical in shape after polymerization. The crosslinking took place because of the free radical polymerization of BMA and NIPAm. Some of PNIPAm can directly graft to the CNF surface by a free radical chain transfer reaction on the polysaccharide substrate.

The SEM images of CNF cryogel microspheres and CNF–PNIPAm hybrid microspheres are shown in Fig. 1. From (a), it is clear that the CNF cryogel microspheres are highly porous, and the morphology of the microspheres appears as a homogeneous network structure. However, under the higher magnification in (b), there are some nanofibril bundles with diameters of approximately hundreds of nanometers. The SEM images (c–f) show that PNIPAm polymer chains were distributed in the CNF network. The morphology of CNF–PNIPAm hybrid microspheres was significantly different from that of CNF cryogel microspheres (a, b). Though the CNF–PNIPAm hybrid microspheres are less porous than pristine CNF cryogel microspheres, the porous structure with different pore sizes is clearly seen on the SEM images. Due to the uneven pore size of CNF cryogel microspheres, the pore size of CNF–PNIPAm hybrid microspheres varied widely from a few nanometers (Cai et al. 2014) to micrometers throughout the cryogel microspheres. Comparing the SEM images of (c–f), it was obvious that when the NIPAm monomer concentration increased during the copolymerization process, the pore size became smaller and the hybrid microspheres became more compact.

The size of the hybrid microspheres was determined by averaging more than 100 particles using an optical microscope using an optical microscope. The diameter of the microspheres was within 20–300 µm with a majority of 50–150 nm, which is shown in Fig. 2.

Chemical structure and thermal stability of CNF–PNIPAm hybrid microspheres

The FTIR spectra of CNF microspheres and CNF–PNIPAm hybrid microspheres were compared in Fig. 3. For pristine CNF, the absorption bands at 3400 and 2900 cm^{-1} are attributed to the stretching of O–H and C–H bonds. The bands at 1636 and 1430 cm^{-1} are due to the bending vibrations of absorbed water and the in-plane bending vibrations of H–CH and O–CH groups, respectively. After the polymerization reaction, the peaks at 1545, 1645 and 1741 cm^{-1} are due to amide groups and N–H groups of NIPAm. The absorption bands of CNF–PNIPAm microspheres from isopropyl groups at 1368 cm^{-1} are also observed. These results indicated that the CNF–PNIPAm hybrid microspheres had been successfully synthesized through radical copolymerization using CNF cryogel microspheres as microreactors.

Thermal stability of the CNF microspheres and CNF–PNIPAm microspheres was studied from the TGA curves. From the DTG curves of CNF microspheres, the maximum decomposition rate occurred at about 323 °C. We can also see the peak of 323 °C in the DTG curves of AM-1 and AM-4. This indicated that the porous structure and high specific area properties remained after copolymerization because the thermal stability of CNF cryogel was significantly different from that of CNF because of its high specific area (Cai et al. 2014). For AM-1 and AM-4, two decomposition rates are observed from the DTG curves; one is 323 °C and the other at about 410 °C. The maximum decomposition rate at 410 °C of AM-4 is much higher than that of AM-1. This reflects that the PNIPAm content in AM-4 is higher than that in AM-1. By comparing the residual content with the weight loss rate of AM-1 and AM-4, the thermal stability of the CNF–PNIPAm microspheres is obviously influenced by the PNIPAm content, probably due to the relatively poor thermal stability of PNIPAm to a large extent (Fig. 4).

Bulk density and flowability of the microspheres

Bulk density was calculated by dividing the weight by the final volume of the sample. The volume was determined by transferring a known quantity of microspheres to a measuring cylinder. The results

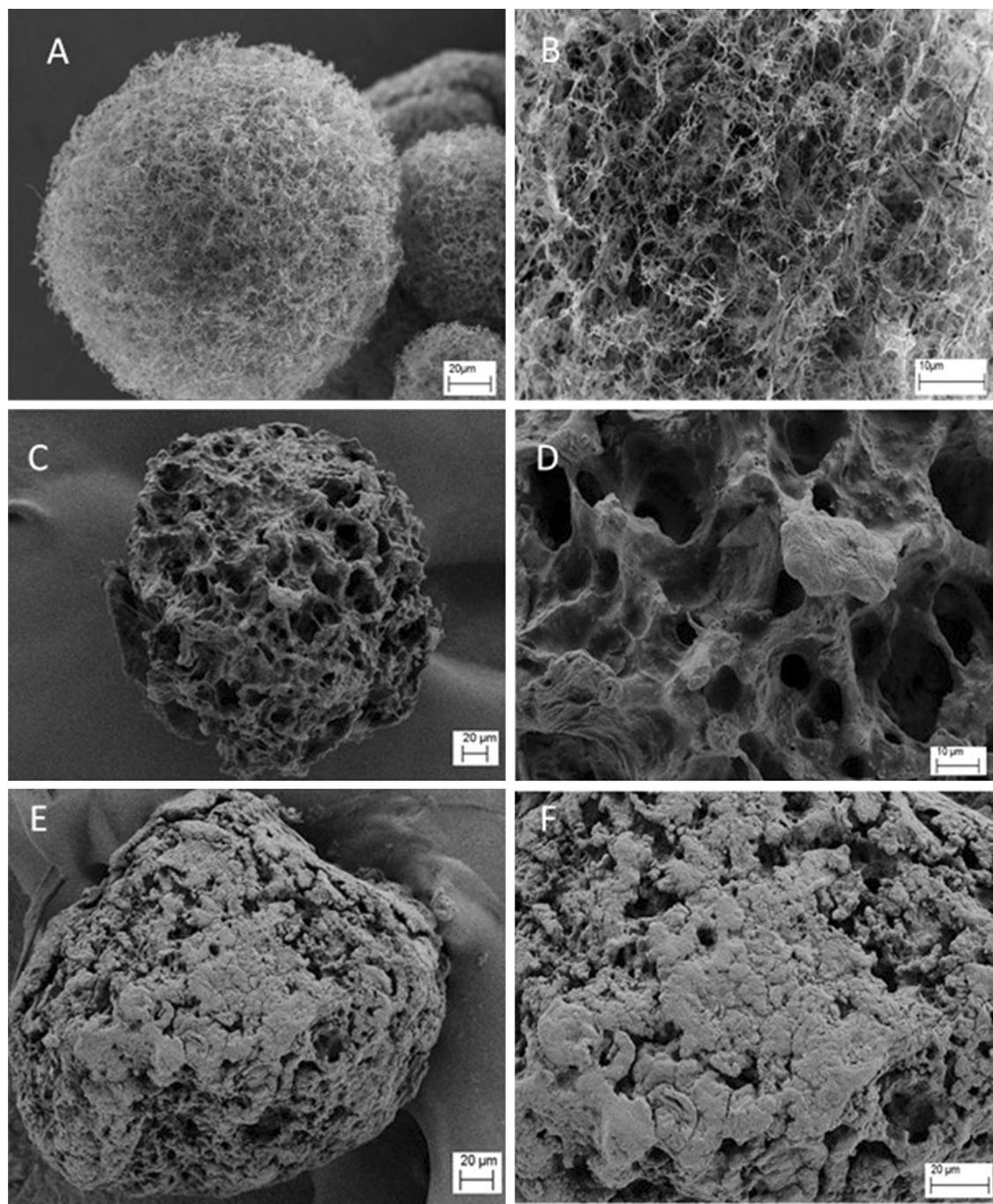


Fig. 1 a, b SEM images of CNF cryogel microspheres; c, d SEM images of AM-1; e, f SEM images of AM-4

are shown in Fig. 5. Generally speaking, the bulk density of different CNF–PNIPAm samples is almost linearly proportional to the PNIPAm content, which indicates that more polymer has been grafted on or entrapped in the microspheres. The angle of repose

of different microspheres was determined by a fixed funnel method. Because CNF microspheres have a very low density, it is hard to measure their angle of repose. The angle of repose of CNF–PNIPAm microspheres was calculated as follows:

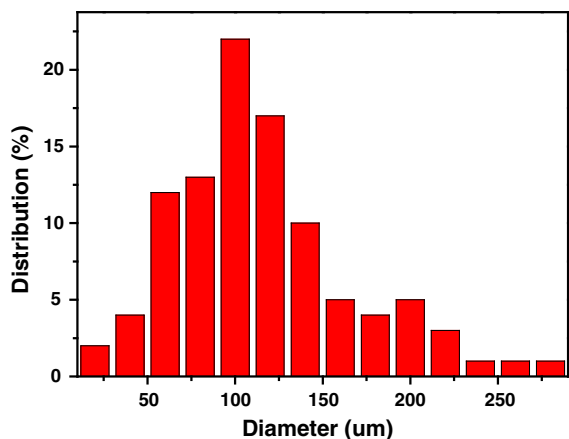


Fig. 2 Particle size distribution (based on number of microspheres) of the CNF–PNIPAm hybrid microspheres

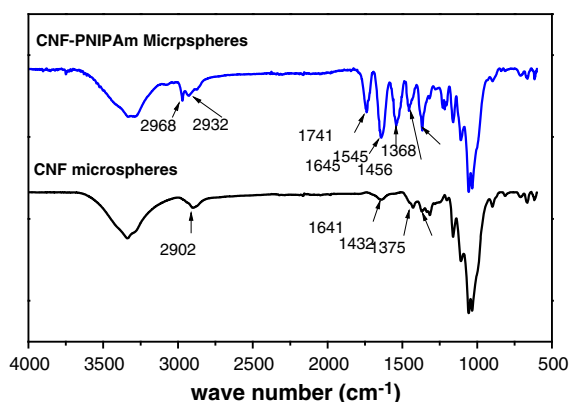


Fig. 3 FTIR of CNF microspheres and CNF–PNIPAm hybrid microspheres

AM-1 = 47.3 ± 5 °C; AM-2 = 44.5 ± 4 °C; AM-3 = 42.4 ± 5 °C; AM-4 = 41.1 ± 6 °C.

Temperature-sensitive water uptake properties of CNF–PNIPAm hybrid microspheres

Figure 6 shows the water uptake behaviors of CNF–PNIPAm microspheres at constant pH 7.2 but varied amounts of PNIPAm and temperature. The water uptake capacity of these CNF–PNIPAm microspheres at 22 °C is in the order of AM-1 > AM-2 > AM-3 > AM-4. This phenomenon is probably due to the different pore structure of those microspheres, which is shown in Fig. 1. The structure of those samples became more compact when increasing the amount of NIPAm monomer, which resulted in lower water

storability. CNF microspheres without incorporating PNIPAm show a high water uptake capacity (about 103 g/g), but they are not temperature-dependent. Figure 6 also indicates that the water uptake capacity of hybrid microspheres tends to decrease at pH 7.2 with increasing temperature for all CNF–PNIPAm samples. The reason is that when the temperature increased from 22 to 37 °C, the shrinkage of PNIPAm in the CNF–PNIPAm microspheres occurred above its LCST. The water absorption property of PNIPAm grafted cellulose has been reported by others as well (Hebeish et al. 2014). Furthermore, the sensitivity of the temperature responsibility of AM-4 is more significant than that of AM-1 under the corresponding temperature, which might be due to the introduction of more PNIPAm chains in the CNF microspheres. However, the difference between AM-3 and AM-4 was not obvious.

In vitro drug release properties

Figure 7 shows the 5-FU release from CNF microspheres and different types of CNF–PNIPAm hybrid microspheres at pH 7.2 and 22 °C. 5-FU, a water-soluble anti-cancer drug, was used as a model drug to study the relationship between release rates from CNF–PNIPAm and CNF microspheres at different temperatures. CNF microspheres incorporated with 5-FU exhibited a much higher release rate compared with other CNF–PNIPAm microspheres. The rapid release of 5-FU from CNF microspheres can be rationalized in terms of the size of the drug and its interactions with CNF microspheres' frameworks. 5-FU is a small, water-soluble molecule, which can easily diffuse out of the micropores of CNF microspheres. The untrapped 5-FU exhibiting a burst effect in PBS within a few seconds has been reported before (Vilac et al. 2013; Lo et al. 2005). In our study, the cumulative amount of released 5-FU reached more than 95 % within a few seconds from CNF microspheres, indicating the drug release rate was almost uncontrollable. However, for CNF–PNIPAm hybrid microspheres, it took more than 1 h to reach a stable release rate. The release amount was only at 94 and 85 % for AM-1 and AM-4 within 3 h. Drug release is controlled by two factors: the diffusivity of the drug and the swelling of the polymer (Bae et al. 1987). Because there are no PNIPAm molecules inside the pristine cryogel microspheres, the model drug

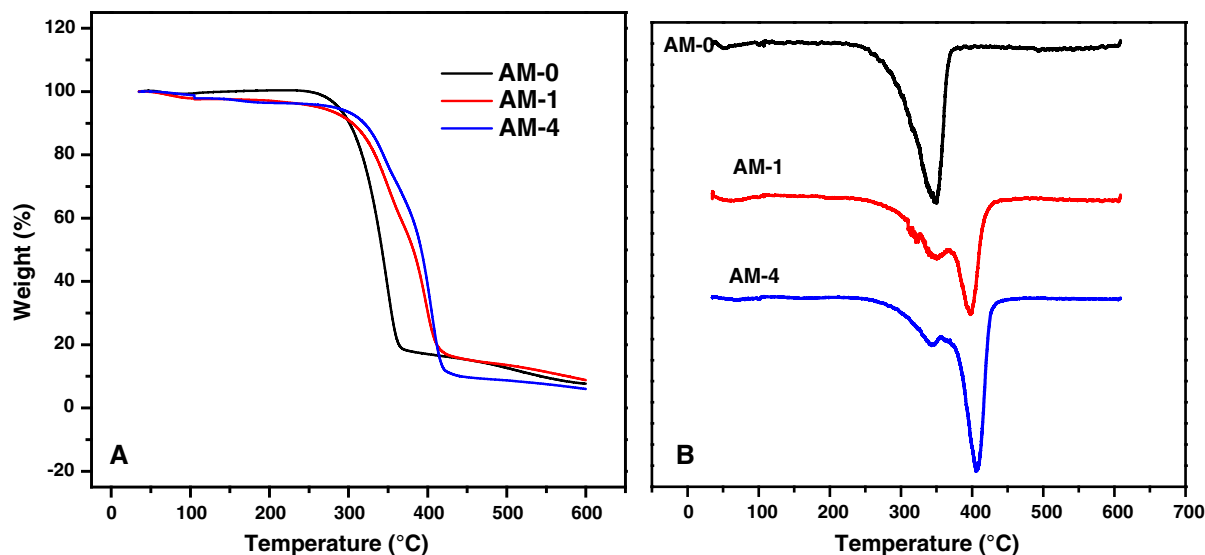


Fig. 4 a TGA and b DTG curves of the CNF microspheres and CNF–PNIPAm hybrid microspheres

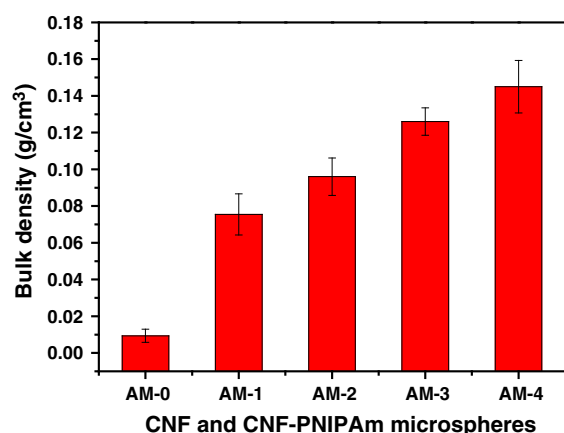


Fig. 5 Bulk density of the CNF microspheres and CNF–PNIPAm hybrid microspheres

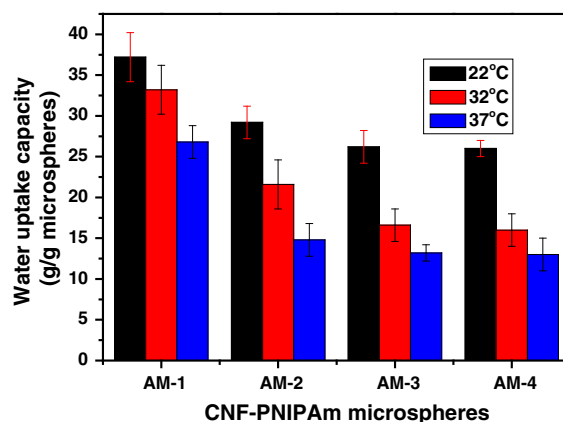


Fig. 6 Temperature-dependent changes of water uptake capacity studies for the CNF–PNIPAm hybrid microspheres

5-FU can diffuse out of the pores of the cryogel microspheres without apparent resistance. CNF cryogel is a typical superabsorbent because of its highly porous structure and hydrophilic nature. The water uptake capacity tests showed that 1 g of the CNF cryogel could absorb more than 100 g of water (Cai et al. 2014). When the drug-loaded CNF microspheres were placed in phosphate buffered saline (PBS), water and 5-FU molecules rapidly diffused into the network. For the CNF–PNIPAm hybrid microspheres, the introduction of PNIPAm polymer chains in the CNF cryogel microspheres would reduce the pore size of

the microspheres, and the swelling ratio would be also reduced, making the drug molecules hard to diffuse into PBS. For different kinds of drug-loaded CNF–PNIPAm microspheres, the release rate decreased with the increase of the PNIPAm amount. These results indicated that the introduction of PNIPAm chains in the CNF microspheres could prolong the drug release and make it controllable.

The release characteristics of 5-FU from CNF–PNIPAm microspheres at different temperatures were also investigated, and the results are shown in Fig. 8. CNF–PNIPAm hybrid microspheres incorporated with

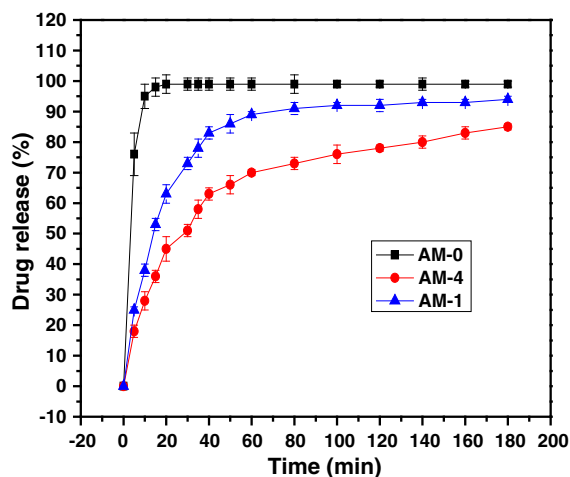


Fig. 7 Effect of the CNF microspheres and CNF-PNIPAm hybrid microspheres on 5-FU release at pH 7.2 measured at 22 °C

5-FU at both 22 °C and 37 °C exhibited a fast release rate pattern up to 60 min, then the release rate flattened out until 3 h. The cumulative amount of released 5-FU by AM-1 and AM-4 reached 89 and 70 % with 60 min at 22 °C, whereas the value was only 66 and 44 % at 37 °C. These significant differences were due to the temperature-sensitive behavior of CNF-PNIPAm hybrid microspheres: when the temperature was above LCST (32 °C), PNIPAm chains aggregated together by the hydrophobic interaction driving force among the hydrophobic groups. The decreased swelling behaviors

and the shrunken pores made the trapped 5-FU molecules difficult to diffuse into PBS. This phenomenon can be also found in other literature (Zhang et al. 2004; Taşdelen et al. 2005). By comparing the release behaviors of AM-1 and AM-4, we can conclude that the amount of PNIPAm in CNF microspheres plays a dominate role in controlling 5-FU release. The different pore structure and swelling properties of those CNF-PNIPAm microspheres could explain this phenomenon.

The fitted equations and values of coefficients of determination for all the release models obtained are as given in Table 1. The 5-FU release kinetic profiles were modeled by fitting the mathematical kinetic models usually used to describe in vitro drug release, including the zero order ($M_t = M_0 + k_0t$), first order ($M_t = M_0e^{-kt}$), Higuchi ($M_t = kt^{1/2}$) and Korsmeyer-Peppas ($M_t/M_\infty = kt^n$) (Costa and Lobo 2001). A comparative study of these values showed that the first-order and Korsmeyer-Peppas models were the best-fit models for all the samples ($R > 0.9$). This is attributed to the swelling of the polymer chains, which leads to the formation of large pores that facilitate drug dissolution into PBS. All the water-soluble drugs generally follow the first-order release mechanism (Risk et al. 1994). The determined values of the diffusion exponent (n) obtained from Korsmeyer-Peppas model ranged between 0.26 and 0.42 (Table 1), indicating that the drug permeation from these 5-FU-loaded microspheres followed Fickian diffusional release (Agnihotri and Aminabhavi, 2006).

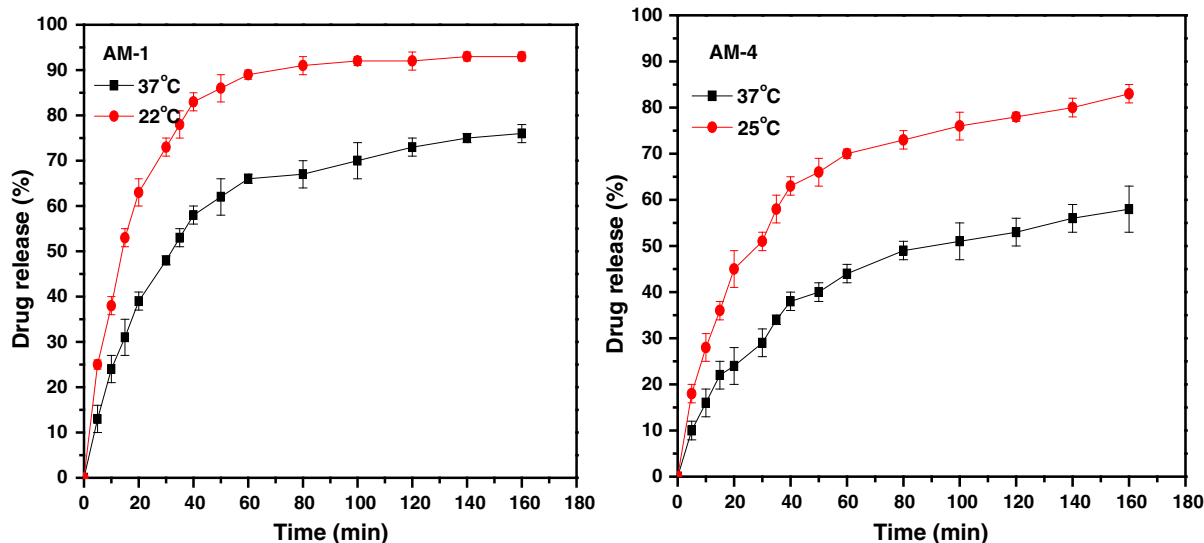


Fig. 8 Effect of CNF-PNIPAm hybrid microspheres on 5-FU release at pH 7.2 measured at 22 and 37 °C in PBS

Table 1 Curve fitting of the mathematical mode of 5-FU loaded CNF–PNIPAM microspheres

Sample-5FU	Zero order	First order	Higuchi	Korsmeyer-Peppas
AM-1 (22 °C)	$M_t = 0.42t + 45.78$ ($R = 0.51994$)	$M_t = 92.31(1-e^{-0.05t})$ ($R = 0.9684$)	$M_t = 6.99y^{1/2} + 8.78$ ($R = 0.7833$)	$M_t = 28.35t^{0.26} - 2.21$ ($R = 0.9039$)
AM-1 (37 °C)	$M_t = 0.39t + 27.59$ ($R = 0.6888$)	$M_t = 73.94(1-e^{-0.037t})$ ($R = 0.9565$)	$M_t = 6.19y^{1/2} + 8.77$ ($R = 0.8987$)	$M_t = 14.03t^{0.36} - 1.88$ ($R = 0.9397$)
AM-4 (22 °C)	$M_t = 0.41t + 31.27$ ($R = 0.6898$)	$M_t = 78.89(1-e^{-0.039t})$ ($R = 0.9409$)	$M_t = 6.4735y^{1/2} + 11.56$ ($R = 0.9037$)	$M_t = 16.49t^{0.33} - 1.61$ ($R = 0.9538$)
AM-4 (37 °C)	$M_t = 0.31t + 16.82$ ($R = 0.8089$)	$M_t = 56.11(1-e^{-0.027t})$ ($R = 0.9584$)	$M_t = 4.74y^{1/2} + 3.13$ ($R = 0.9651$)	$M_t = 7.49t^{0.42} - 0.91$ ($R = 0.9775$)

Conclusion

In summary, we successfully fabricated thermal-sensitive CNF–PNIPAm hybrid cryogel microspheres by using native cellulose nanofibril cryogel microspheres as the microreactor. The robust and highly porous structure of CNF microspheres allowed the CNF–PNIPAm microspheres to have good stability. Also, the CNF–PNIPAm hybrid microspheres were sensitive to the temperature environment, and the water swelling behaviors changed greatly by varying the temperature from 22 to 37 °C. Moreover, the porous system and physicochemical properties of CNF–PNIPAm could be controlled by changing the concentration of the NIPAm aqueous solution during the preparation process. The drug release rate in PBS from different CNF–PNIPAm hybrid microsphere samples at 22 and 37 °C changed significantly. Hence, the thermal-responsive CNF–PNIPAm microspheres can be used as an excellent drug delivery device with controllable drug release characteristics. In addition, this synthesis technique using natural porous spherical cryogel as a microreactor to prepare hybrid composite porous material could be extended to a variety of other systems in order to fabricate functional porous materials.

References

- Agnihotri SA, Aminabhavi TM (2006) Novel interpenetrating network chitosan-poly(ethylene oxide-g-acrylamide) hydrogel microspheres for the controlled release of capcitabine. *Int J Pharm* 324:103–115
- Ayers MR, Hunt AJ (2001) Synthesis and properties of chitosan–silica hybrid aerogels. *J Non-Cryst Solids* 285:123–127
- Bae YH, Okano T, Hsu R (1987) Thermo-sensitive polymers as on-off switches for drug release. *Die Makromol Chem Rapid Commun* 8:481–485
- Cai H, Sharma S, Liu W, Mu W, Liu W, Zhang X, Deng Y (2014) Aerogel microspheres from natural cellulose nanofibrils and their application as cell culture scaffold. *Biomacromolecules* 15:2540–2547
- Costa P, Lobo JMS (2001) Modeling and comparison of dissolution profiles. *Eur J Pharm Sci* 13:123–133
- García-González CA, Alnaief M, Smirnova I (2011) Polysaccharide-based aerogels-promising biodegradable carriers for drug delivery systems. *Carbohydr Polym* 86:1425–1438
- Gavillon R, Budtova T (2007) Aerocellulose: new highly porous cellulose prepared from cellulose-NaOH aqueous solutions. *Biomacromolecules* 9:269–277

- Guenther U, Smirnova I, Neubert RHH (2008) Hydrophilic silica aerogels as dermal drug delivery systems—Dithranol as a model drug. *Eur J Pharm Biopharm* 69:935–942
- Hebeish A, Farag S, Sharaf S, Shaheen TI (2014) Thermal responsive hydrogels based on semi interpenetrating network of poly (NIPAm) and cellulose nanowhiskers. *Carbohydr Polym* 102:159–166
- Innerlohinger J, Weber HK, Kraft G (2006) Aerocellulose: aerogels and aerogel-like materials made from cellulose. *Macromol Symp* 244:126–135
- Kadib AE, Molvinger K, Guimon C, Quignard F, Brunel D (2008) Design of stable nanoporous hybrid chitosan/titania as cooperative bifunctional catalysts. *Chem Mater* 20:2198–2204
- Litschauer M, Neouze MA, Haimer E, Henniges U, Potthast A, Rosenau T, Liebner F (2011) Silica modified cellulosic aerogels. *Cellulose* 18:143–149
- Lo CL, Lin KM, Hsiue GH (2005) Preparation and characterization of intelligent core-shell nanoparticles based on poly(D, L-lactide)-g-poly(N-isopropyl acrylamide-co-methacrylic acid). *J Controlled Release* 104:477–488
- Mehling T, Smirnova I, Guenther U, Neubert RHH (2009) Polysaccharide-based aerogels as drug carriers. *J Non-Cryst Solids* 355:2472–2479
- Mikkonen KS, Parikka K, Ghafar A, Tenkanen M (2013) Prospects of polysaccharide aerogels as modern advanced food materials. *Trends Food Sci Tech* 34:124–136
- Molvinger K, Quignard F, Brunel D, Boissiere M, Devoisselle J (2004) Porous chitosan-silica hybrid microspheres as a potential catalyst. *Chem Mater* 16:3367–3372
- Novotna K, Havelka P, Sopuch T, Kolarova K, Vosmanska V, Lisa V, Svorcik V, Bacakova L (2013) Cellulose-based materials as scaffolds for tissue engineering. *Cellulose* 20:2263–2278
- Risk S, Duru D, Gaudy D, Jacob M (1994) Natural polymer hydrophilic matrix: influencing drug release factors. *Drug Dev Ind Pharm* 16:2563–2574
- Ritschel WA, Thompson GA, Lucker PW, Wetzelsberger K (1980) Biopharmaceutic evaluation of etofylline clofibrate and its drug formulation. *Arzneim Forsch Drug Res* 30:2020–2023
- Sai H, Fu R, Xing L, Xiang J, Li Z, Li F, Zhang T (2015) Surface modification of bacterial cellulose aerogels' web-like skeleton for oil/water separation. *ACS Appl Mater Interface* 7:7373–7381
- Smirnova I, Mamic J, Arlt W (2003) Adsorption of drugs on silica aerogels. *Langmuir* 19:8521–8525
- Song W, Lima AC, Mano JF (2010) Bioinspired methodology to fabricate hydrogel spheres for multi-applications using superhydrophobic substrates. *Soft Matter* 6:5868–5871
- Taşdelen B, Kayaman-Apohan N, Güven O, Baysal BM (2005) Preparation of poly (N-isopropylacrylamide/itaconic acid) copolymeric hydrogels and their drug release behavior. *Radia Phys Chem* 73:340–345
- Ulker Z, Erkey C (2014) An emerging platform for drug delivery: aerogel based systems. *J Control Release* 177:51–63
- Vilac N, Amorim R, Machado AF, Parpot P, Pereira MFR, Sardo M, Rocha J, Fonseca AM, Neves IC, Baltazar F (2013) Potentiation of 5-fluorouracil encapsulated in zeolites as drug delivery systems for in vitro models of colorectal carcinoma. *Colloids Surf B* 112:237–244
- You YZ, Kalebaila KK, Brock SL (2008) Temperature-controlled uptake and release in PNIPAM-modified porous silica nanoparticles. *Chem Mater* 20:3354–3359
- Zhang JT, Huang SW, Cheng SX, Zhuo RX (2004) Preparation and properties of poly (N-isopropylacrylamide)/poly (N-isopropylacrylamide) interpenetrating polymer networks for drug delivery. *J Polym Sci Pol Chem* 42:1249–1254
- Zhang W, Zhang Y, Lu C, Deng Y (2012) Aerogels from crosslinked cellulose nano/micro-fibrils and their fast shape recovery property in water. *J Mater Chem* 22:11642–11650
- Zhang X, Lin Z, Chen B, Zhang W, Sharma S, Gu W, Deng Y (2014) Solid-state flexible polyaniline/silver cellulose nanofibrils aerogel supercapacitors. *J Power Sources* 246:283–289